IGCSE 0607 Practice Questions — E2.2

Algebraic manipulation

Total Questions: 30 — **Total Marks:** 168 Marks

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- **E2.2** Algebraic manipulation
  → **Paper 2 (Non-Calculator)** *(24 Qs)* · **Paper 4 (Calculator)** *(6 Qs)*

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1 Expand and simplify.

$(3x - 2)(2x + 5)$

..... **[2]**

2 **(a)** Expand and simplify.

$3(2x - 1) - 2(x + 4)$

..... **[2]**

(b) Factorise fully.

$15x^2y - 20xy^2$

..... **[2]**

3 **(a)** Expand and simplify.

$4(2a - 3b) - 3(a - 5b)$

..... **[2]**

(b) Factorise completely.

$12x^2y - 18xy^2$

..... **[2]**

4 **(a)** Expand and simplify.

$4(2x - 3y) - 3(x - 5y)$

..... **[2]**

(b) Factorise completely.

$14a^2b - 21ab^2$

..... **[2]**

5 **(a)** Expand and simplify

$4(3x - 2) - 2(x - 5)$

..... **[2]**

(b) Factorise fully

$15a^2b - 10ab^2$

..... **[2]**

6 **(a)** Expand and simplify.

$3(2x - 5) - 2(x + 4)$

..... **[2]**

(b) Factorise completely.

$8x^2y - 12xy^2$

..... **[2]**

7 **(a)** Simplify

$$4x^2 - 7xy + 3 - 9x^2 + 2xy - 8$$

..... **[2]**

(b) Factorise fully

$$14a^3b - 21a^2b^2$$

..... **[2]**

8 **(a)** Expand and simplify

$$3(2x - 5) - 4(x - 2)$$

..... **[2]**

(b) Factorise fully

$$12x^2y - 18xy^2$$

..... **[2]**

9 **(a)** Simplify.

$3a - 5b + 2a + 8b$

..... **[2]**

(b) Factorise completely.

$12x^2y - 18xy^2$

..... **[2]**

10 **(a)** Expand and simplify.

$3(2x - 5) - 4(x - 3)$

..... **[2]**

(b) Factorise fully.

$14x^2y + 21xy^2$

..... **[2]**

11 **(a)** Factorise completely.

$12x^2 - 27y^2$

..... **[2]**

(b) Factorise completely.

$2ax - 6ay + bx - 3by$

..... **[2]**

12 **(a)** Expand and simplify.

$(2x - 3)(x + 5)$

..... **[2]**

(b) Factorise.

$y^2 - 16x^2$

..... **[1]**

(c) Factorise.

$p^2 - 4pq - 5q^2$

..... **[2]**

13 **(a)** Expand and simplify.

$(2x - 3)(x + 5)$

..... **[2]**

(b) Factorise.

$ac - 3bc + 2ad - 6bd$

..... **[2]**

(c) Factorise.

$16x^2 - 25y^2$

..... **[1]**

14 **(a)** Expand and simplify.

$(2x - 3)(x + 4)$

..... **[2]**

(b) Factorise completely.

$ax - bx - 3ay + 3by$

..... **[2]**

(c) Factorise.

$49x^2 - 16y^2$

..... **[2]**

15 **(a)** Factorise.

$x^2 - 8x + 15$

..... **[2]**

(b) Factorise.

$2x^2 - 5x - 12$

..... **[2]**

(c) Factorise.

$ax - 3a + 2bx - 6b$

..... **[2]**

16 **(a)** Expand and simplify

$$(3x-2)(x+5)-(2x-3)^2$$

..... **[3]**

(b) Factorise completely:

(i) $3x^3-12xy^2$

..... **[2]**

(ii) $6x^2-11x-10$

..... **[2]**

17 **(a)** Expand and simplify.

$(2x - 5)(3x + 1)$

..... **[2]**

(b) Factorise.

$x^2 - 7x - 18$

..... **[2]**

(c) Factorise completely.

$18x^2 - 50y^2$

..... **[3]**

18 **(a)** Expand and simplify.

$(x-2)(2x+3)(x-5)$

..... **[3]**

(b) Factorise completely.

$12x^3 + 17x^2 - 5x$

..... **[3]**

19 **(a)** Factorise completely.

$18x^2 - 50y^2$

..... **[3]**

(b) Factorise completely.

$3x^3 + 10x^2 - 8x$

..... **[3]**

20 **(a)** Expand and simplify.

$$(x-2)(x+3)(2x-1)$$

..... **[3]**

(b) Factorise completely.

(i)

$$6x^2 - 11x - 10$$

..... **[2]**

(ii)

$$ax - ay + 3cx - 3cy$$

..... **[2]**

21 **(a)** Factorise

$$cx - 5dx + 3cy - 15dy$$

..... **[2]**

(b) Factorise fully

$$2x^3 - 18x$$

..... **[2]**

(c) Factorise fully

$$12x^3 - 14x^2 - 10x$$

..... **[3]**

22 **(a)** Expand and simplify.

$(x-4)(2x+1)(x+3)$

..... **[3]**

(b) Factorise completely.

$2ax - 3by + 2ay - 3bx$

..... **[2]**

(c) Factorise.

$6x^2 + 7x - 20$

..... **[2]**

23 **(a)** Expand and simplify

$$(2x-1)(x+3)(3x-2)$$

..... **[3]**

(b) Factorise completely

$$4ax - 6ay - 6bx + 9by$$

..... **[2]**

(c) Given that $x^2 - y^2 = 45$ and $x - y = 5$, find the value of $(x+y)^2$.

..... **[3]**

24 **(a)** Factorise

$$6ax - 2ay - 3bx + by$$

..... **[2]**

(b) Factorise completely.

(i)

$$50x^2 - 8y^2$$

..... **[3]**

(ii)

$$2x^3 + 5x^2 - 12x$$

..... **[3]**

25 **(a)** Expand and simplify

$$(x-3)(2x+1)(x+4)$$

..... **[3]**

(b) Factorise

$$2x^2 - 7x - 15$$

..... **[2]**

26 **(a)** Expand and simplify.

$(2x - 3)(x + 5)$

..... **[2]**

(b) Factorise.

$x^2 - 7x - 18$

..... **[2]**

(c) Factorise completely.

$4x^2 - 25y^2$

..... **[2]**

27 **(a)** Expand and simplify

$$(x-4)(2x+3)(x+2)$$

..... **[3]**

(b) Factorise

$$16x^2 - 49y^2$$

..... **[2]**

(c) Factorise

$$3x^2 - 10x - 8$$

..... **[2]**

28 **(a)** Expand and simplify.

$(x-4)(2x+3)(x+1)$

..... **[3]**

(b) Factorise completely.

$6x^2 - 11x - 10$

..... **[2]**

(c) Factorise completely.

$x^3 + 3x^2 - 4x - 12$

..... **[3]**

29 **(a)** Expand and simplify.

$(x-2)(2x+1)(x+3)$

..... **[3]**

(b) Factorise.

$6x^2 - 11x - 10$

..... **[2]**

(c) Factorise completely.

$3x^3 + 5x^2 - 2x$

..... **[3]**

30 **(a)** Expand and simplify.

$(x-2)(x+3)(2x-1)$

..... **[3]**

(b) Factorise.

$6x^2 - 11x - 10$

..... **[2]**

(c) Factorise completely.

$2x^3 - 18x$

..... **[3]**